WDC - Western Digital | Full Deep Dive

Full investment research | Model assumptions and source evidence

October 8, 2026 input update | Framework 8.0 | NOT HELD | BOTTLENECK - qualified nearline HDD capacity | High Conviction

Analytical Base: 46/60 (77%). Valuation: 20x forward four-quarter adjusted operating profit.

Snapshot: USD 448.17 per share on September 21, 2026 close; 360.541 million shares, implying USD 161,583.63 million equity value. Source [WDC-6]. Price is a dated header snapshot, not a valuation input in the body.

Active Q4 2027 ceiling: USD 800.93/share. Q1 2028: USD 853.97/share.

All financial model amounts are USD millions, shares in millions, unless stated. Calendar buckets map to the nearest fiscal quarter: Q3 2026 = FQ1 FY2027, Q4 = FQ2, Q1 2027 = FQ3 and Q2 = FQ4. Fiscal year ends around June; July 3 actuals belong to the June/Q2 bucket. The fiscal mapping is shown for every modeled quarter.

Spot-market scarcity and realized pricing

Morgan Stanley channel checks report recently closed **non-contracted nearline HDD deals at 3-5 cents/GB, or $30-50/TB**, with some higher observations. An exact-match secondary synopsis dates the note to October 5, 2026; the supplied crop itself has no visible date or author header. The original deal sample, product mix, geography, buyer size and payment terms are absent. These are attributed channel transactions, not company-reported blended ASPs or booked hyperscaler contract rates.

The same crop recalls June checks that suppliers sought **2.5-3 cents/GB ($25-30/TB) blended nearline pricing by CY2027/CY2028**, versus a cited 1.5 cents/GB comparison. Those are vendor aspirations reported by the broker. They are neither an issued price list nor a second demand event. The broker's 300 EB supply gap in 2026 and 400 EB in 2027/2028 are unmet-demand estimates, not saleable output. Broader neocloud, agentic and physical-AI retention reinforces the existing demand thesis; no additional exabyte volume is added.

Bounded commercial case

Assign **30% probability** that a limited part of qualified shipments realizes the pricing opportunity. Use the **$30/TB low endpoint** and expose **10% of already modeled, qualified shipped exabytes**. For each quarter, only the positive gap between $30 and the existing reference ASP is eligible. This subtraction removes the pricing already included in the model. The unexposed 90% captures fixed contracts, customer bargaining power and the limited reach of the observed spot sample. This exposure is an analytical bound, not a disclosed spot-sales mix.

The 30% weight reflects tangible reported transactions but limited evidence that they transmit to these companies' blended realizations. The realization phase is 25% in Q4 2026, 50% in Q1 2027, 75% in Q2 and 100% from Q3 2027 through Q4 2028. It fades to 75%, 50%, 25% and zero across 2029, remaining zero thereafter. This is a timing assumption for shipment and acceptance of new or repriceable orders. No premium is applied to Q3 2026. Repeated broker or social references receive full overlap with this one branch.

Weighted ASP increment = 30% × 10% × phase × MAX(0, $30 - reference ASP). Incremental revenue = already qualified shipped EB × weighted ASP increment. With EB and $/TB, the result is USD millions. Incremental revenue carries **10% COGS** for support, fulfillment and component cost and **3% operating expense**, leaving **87% incremental adjusted operating profit**. The COGS increment is allocated among HDD data-center products in proportion to their modeled revenue. Existing

production costs, R&D and SG&A remain. The 3% increment enters other opex. These are explicit bounded cost assumptions, not issuer marginal-cost guidance.

Active ASP equals the reference ASP, less the established **absolute Toshiba price concession**, plus this scarcity increment. The Toshiba dollar loss is not recalculated on the higher ASP. This prevents a new interaction penalty or double counting the two branches. Volumes, density, qualification factors, product mix, diluted-share schedule, the 20× F4Q adjusted-operating-profit method and Base score remain unchanged. The pricing evidence updates operating economics; it does not establish a new durable monopoly or warrant another score point.

Additionally, our checks uncovered important HDD demand and pricing-related updates. First, HDD demand has accelerated in the last 1-2 months - the core underlying driver of this acceleration wasn't pinned on any one specific workload or factor, but rather speaks to the growing criticality of data generation/retention as new AI workloads emerge. We believe agentic AI could represent one emerging incremental driver - relative to chatbot workloads, agents create more persistent data and context that must be retained across interactions, increasing storage intensity for HDDs. Second, demand is also broadening as storage capacity at CSPs is limited and NAND prices remain 17-20x HDD prices - for example, neoclouds have emerged as new buyers of HDDs (historically they relied on the CSPs). As have humanoid/physical AI companies. However, it seems HDD procurement is very challenging for this cohort given supply is still tracking ~300EB's below demand this year, and expanding to a 400EB gap in CY27/CY28 (this is helping tape too, FWIW). Lastly, pricing remains a key upside revenue/margin tailwind. Recall back in June we highlighted checks indicating the HDD vendors were targeting blended nearline pricing at 2.5-3c per gigabyte (vs. 1.5c/GB today) by CY27/CY28. Recently, we have heard about non-contracted nearline deals getting closed at 3-5c per GB, or $30-50 per terabyte, with even some examples of upside to these prices. While we can't yet extrapolate the breadth of these spot rates across the full nearline market, hearing about actual transactions clear 3–5c/GB provides tangible validation that the pricing assumptions underpinning our bull-case EPS are increasingly achievable.

Original Morgan Stanley HDD channel-check crop. Likely October 5, 2026 by matching dated synopsis; crop itself undated. Spot deals, blended supplier aspirations and unmet demand use distinct scopes.

Quarterly pricing and cost bridge

Quarter	Shipped EB	Reference $/TB	Phase	Weighted $/TB gain	Revenue gain $M	COGS $M	Opex $M	OP gain $M
Q3 2026	223.000	16.368	0.00%	0.000	0.000	0.000	0.000	0.000
Q4 2026	241.000	16.777	25.00%	0.099	23.901	2.390	0.717	20.794
Q1 2027	260.370	17.196	50.00%	0.192	50.005	5.001	1.500	43.505
Q2 2027	279.870	17.540	75.00%	0.280	78.460	7.846	2.354	68.260
Q3 2027	294.378	17.891	100.00%	0.363	106.938	10.694	3.208	93.036
Q4 2027	312.435	18.159	100.00%	0.355	110.982	11.098	3.329	96.555
Q1 2028	326.625	18.159	100.00%	0.355	116.023	11.602	3.481	100.940
Q2 2028	342.495	18.159	100.00%	0.355	121.660	12.166	3.650	105.844
Q3 2028	363.825	18.159	100.00%	0.355	129.237	12.924	3.877	112.436
Q4 2028	373.245	18.159	100.00%	0.355	132.583	13.258	3.977	115.347
Q1 2029	404.920	18.114	75.00%	0.267	108.291	10.829	3.249	94.213
Q2 2029	425.484	18.068	50.00%	0.179	76.151	7.615	2.285	66.251
Q3 2029	440.440	18.023	25.00%	0.090	39.563	3.956	1.187	34.420
Q4 2029	440.209	17.978	0.00%	0.000	0.000	0.000	0.000	0.000
Q1 2030	487.552	17.887	0.00%	0.000	0.000	0.000	0.000	0.000
Q2 2030	510.272	17.797	0.00%	0.000	0.000	0.000	0.000	0.000
Q3 2030	532.980	17.707	0.00%	0.000	0.000	0.000	0.000	0.000
Q4 2030	539.536	17.618	0.00%	0.000	0.000	0.000	0.000	0.000

The Q4 2027-Q3 2028 valuation window contains **$477.902142M incremental revenue and $415.774863M incremental adjusted OP**. Total active F4Q OP is **$15,538.087593M**, valued at 20× with 388.000M diluted shares, yielding **$800.932350/share**. Analytical and portfolio Excel Base are both **77/100**. These current outputs include the selected branch once.

Counter-evidence and next checks: fixed-volume/price LTAs can insulate most shipments; a few scarce-drive transactions may include distributor margins or urgent delivery premiums that do not reach manufacturers. Drive density, Toshiba expansion, improved yields or customer inventory digestion can close the shortage. Recheck realized $/TB, shipment mix, new-contract terms, customer acceptance and incremental component/support cost at the next issuer results and through 2027 renewals. Lower the branch if spot transactions fail to reach new-order realizations; increase exposure only with a defined qualified, accepted shipment pool. The NAND/HDD 17-20× comparison has unresolved product and contract scope and supplies no separate model input.

Dated secondary synopsis used solely to reconstruct source identity. The image itself is the supplied evidence; the synopsis is not independent confirmation of transaction economics.

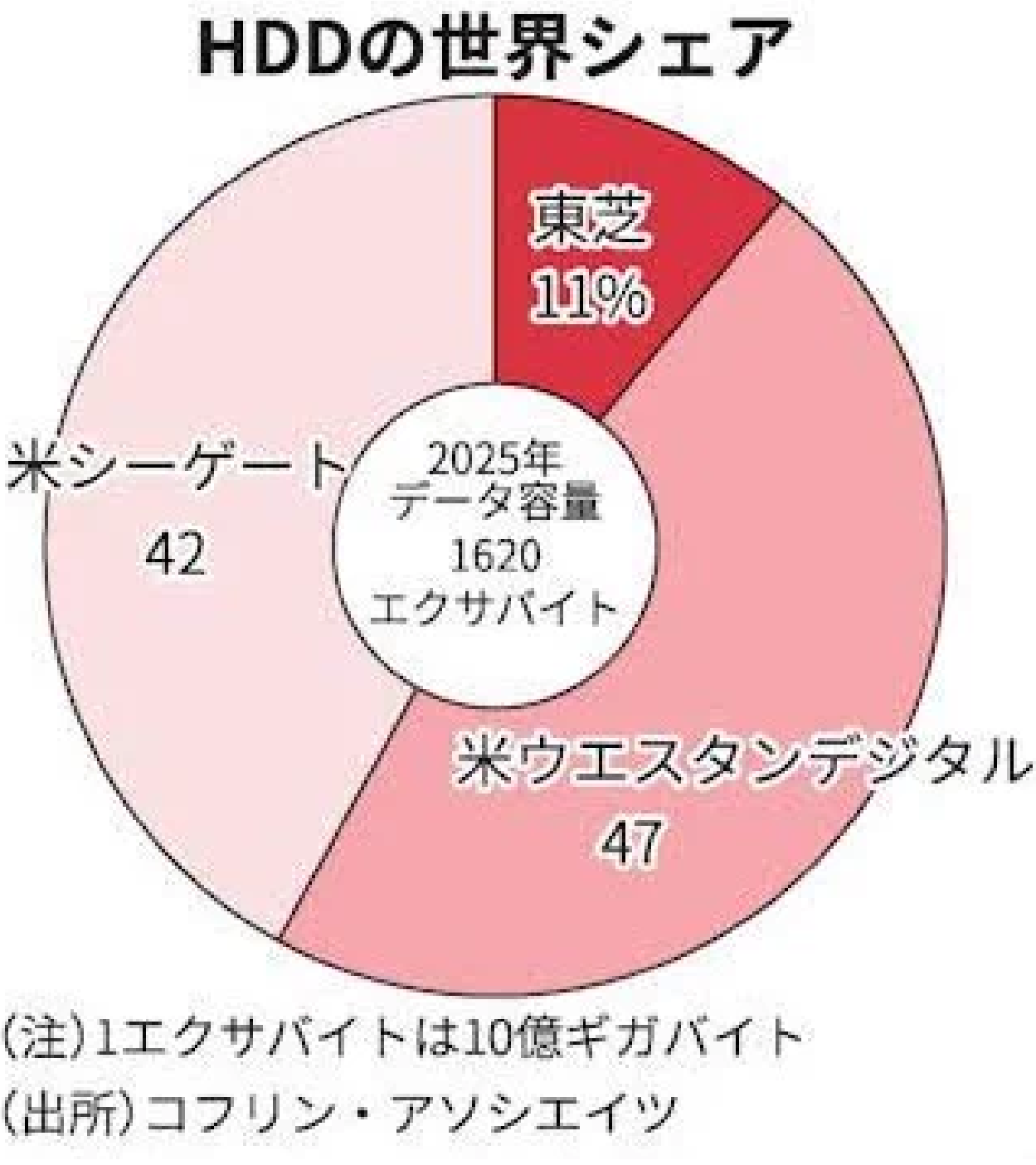
Qualified competitor supply and the pricing channel

Toshiba's October 2 issuer release says the expanded Laguna Technopark nearline HDD line has made its first shipments. TIP targets nearly twice its FY2025 annual production by FY2027 **on a storage-capacity basis**. This measures bytes, not drive units, and Toshiba FY2027 ends in March 2028. Exact monthly accepted exabytes, utilization and ASP are undisclosed. Density gains are not added again to the stated capacity target.

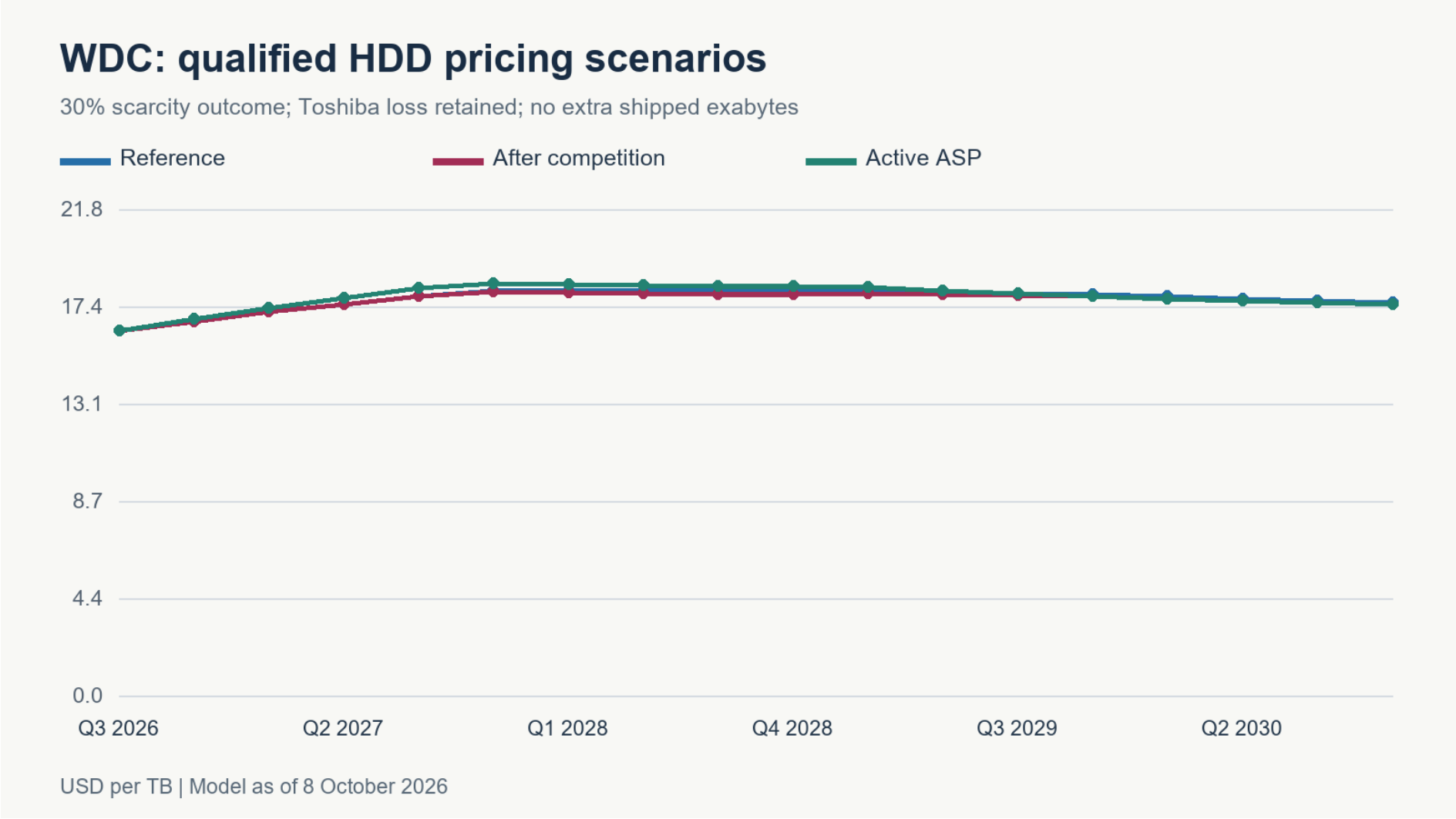
The existing competitor branch assigns **50% probability to a 4% concession on 50% of new or repriceable cloud/data-center revenue**. Its full-ramp probability-weighted reduction is 1% of the reference price. The phase is 25% in Q4 2027, 50% in Q1 2028, 75% in Q2 and full from Q3 2028. From 2029, deduct 50% overlap with normalization already in the reference. Initial shipments establish an execution milestone, not immediate renewal of every customer contract. Demand absorption, qualification and protected contracts limit the economic outcome.

This absolute price/OP loss is retained alongside the separate scarcity branch. At unchanged volume, a price concession does not remove production costs. No extra unit loss is assigned. A later technology expansion discussed for 2030 must not erase the issuer's initial-shipment milestone or be conflated with the same capacity denominator. Competition remains a scored risk.

Toshiba original release, October 2. Reposts describe the same event. The capacity target is not proof of full output.



Nikkei/Techno Systems Research exhibit supplied in the DOT digest: 2025 HDD share by storage capacity, not units or revenue. Toshiba 11%, Western Digital 47%, Seagate 42%; total 1,620 EB. The chart does not quantify the new factory output.



WDC: qualified HDD pricing scenarios. USD per TB. 30% scarcity outcome; Toshiba loss retained; no extra shipped exabytes

Quarter	Reference ASP	Toshiba-only ASP	Active ASP	Toshiba OP loss $M
Q3 2026	16.368	16.368	16.368	0.000
Q4 2026	16.777	16.777	16.876	0.000
Q1 2027	17.196	17.196	17.388	0.000

Quarter	Reference ASP	Toshiba-only ASP	Active ASP	Toshiba OP loss $M
Q2 2027	17.540	17.540	17.821	0.000
Q3 2027	17.891	17.891	18.254	0.000
Q4 2027	18.159	18.114	18.469	-14.184
Q1 2028	18.159	18.069	18.424	-29.657
Q2 2028	18.159	18.023	18.378	-46.646
Q3 2028	18.159	17.978	18.333	-66.069
Q4 2028	18.159	17.978	18.333	-67.779
Q1 2029	18.114	18.023	18.291	-36.673
Q2 2029	18.068	17.978	18.157	-38.439
Q3 2029	18.023	17.933	18.023	-39.690
Q4 2029	17.978	17.888	17.888	-39.570
Q1 2030	17.887	17.798	17.798	-43.605
Q2 2030	17.797	17.708	17.708	-45.407
Q3 2030	17.707	17.619	17.619	-47.189
Q4 2030	17.618	17.530	17.530	-47.528

1. Thesis, bottleneck and expanded TAM

Western Digital is a standalone HDD company after the February 2025 SanDisk separation. The opportunity is capacity storage for cloud and AI: data accumulates as compute is reused, and HDDs provide an economical persistent tier. The company sells heads/media engineering, manufacturing yield, qualified reliability and capacity per drive. It is not a NAND producer in this model. [WDC-2]

Scarcity rests on a small qualified supplier base and long manufacturing and validation cycles. Higher platter density expands exabytes without a proportional increase in finished-drive units. Western Digital can monetize its current ePMR/UltraSMR platform while preparing HAMR, limiting the need for an abrupt migration. This dual path is an execution choice, not proof of a permanent monopoly.

The next opportunities are persistent inference context, video, sovereign/enterprise clouds and physical-AI data. A WD-sponsored IDC survey found 61% of surveyed organizations had AI-related data growth of at least 25% in the past year. This is supporting demand evidence, not an industry revenue TAM or a booking. The model treats the installed-data effect within a capped exabyte demand bridge. [WDC-8]

Seagate is the primary scaled peer and already has commercial HAMR experience; Toshiba is the other major HDD producer. SSD/NAND suppliers compete in high-performance tiers and at the storage boundary. HDD supply discipline can break if density or capacity expands faster than demand, even without a new entrant. No microcap direct substitute for a fully qualified nearline HDD platform was identified. UltraSMR is a recording/data-placement approach that may coexist with ePMR or HAMR; their market shares cannot be added as separate TAMs. [WDC-2, WDC-3]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
2026-27: present storage wallet	Cloud remains the dominant reported end market; 40TB ePMR enters volume	A capacity and pricing step inside the existing HDD revenue pool.
2027-28: HAMR and performance	44TB qualification, then higher density and high-bandwidth functions	Probability-weighted HAMR shipments; all replace part of older capacity.
2029-30+: compounding retained data	Wider enterprise, physical AI and larger data lakes	Retention demand index; no separate robotics or all-flash displacement TAM added.

AH-12 / Plan-B demand bridge: customer identities and supplier wallet shares are confidential. Amazon, Microsoft, Alphabet and Meta form a representative spending basket, not a claim that each is a disclosed Western Digital customer. Their latest CY2026 capex figures are $220B, $175B, $200B midpoint and $137.5B midpoint respectively, totaling $732.5B [D1-D4]. Definitions differ: Amazon is cash capex, Meta includes finance-lease principal, and Microsoft changed lease classification; the basket is a demand index, not HDD TAM. Memory inflation is not extra physical demand. Calibrate one effective supplier coefficient against the latest actual quarter: $3,335M / ($732,500M / 4) = 1.82116%. Applying it to the quarterly run-rate basket reproduces $3,335M exactly (0.0% calibration error). This coefficient embeds non-basket cloud/enterprise business; individual customer shares are NOT separately knowable. Calibration of one point does not establish forecast accuracy. Each customer is extended at the same analyst-assumed capex growth: +25% in 2027, +20% in 2028, +14% in 2029 and +10% in 2030. Thus basket capex is $915.625B, $1,098.750B, $1,252.575B and $1,377.833B. Quarterly allocation is one quarter of each annual total, a normalization rather than a forecast of actual quarterly cash spending. Demand EB = 207.9 anchor EB x (basket capex / $732.5B) x retained-data intensity. Intensity equals 1.15 / 1.20 in Q3/Q4 2026, 1.25 in 2027-28, 1.30 in 2029 and 1.45 in 2030; these LOW-confidence analyst indices represent accumulating installed-base data. Potential EB = MIN(demand EB, manufacturing capacity EB). Apply the separate HAMR qualification reserve to determine shipped EB. Revenue then equals shipped EB x modeled USD/TB, with no second capex multiplier. Capacity, demand, mix and ASP are separately visible below. This is a calibrated proxy, not a company-disclosed customer model. The company reports 231 total EB in the latest quarter, not a precise cloud-only EB figure in the source set. The model allocates 90% to cloud (207.9 EB) as a LOW-confidence physical proxy; $3,335M cloud revenue is independently anchored to the reported $3,747M total less $225M client and $187M consumer. Changing that EB split changes implied USD/TB, not the calibrated actual cloud revenue.

2. Latest evidence and financial anchor

Latest two earnings calls reviewed: April 30 and August 5, 2026, including management answers. April established strong cloud demand, four HAMR qualifications, long-duration customer planning and continuing cost-per-TB improvements. Its then-current June guidance is superseded by the July 3 actuals in the August release. [WDC-1, WDC-3, WDC-4]

August management placed 40TB ePMR into volume ramp at two customers, targeted at least half of nearline exabytes on that platform by FQ3 FY2027, and expected UltraSMR to reach about 60% by fiscal year-end. It described 44TB HAMR timing in H1 2027 and high-bandwidth samples at five customers. Long-range LTA

discussions into 2031 are not the same as fully priced, binding orders for that entire period. [WDC-3]

The September 9 update sharpened timing: small HAMR shipments in Q1 2027, more in Q2, then a broader H2 ramp. It also identified DRAM/NAND input-cost inflation and indicated capex could run slightly above the long-term 4-6% range over the next couple of years. Those costs matter even though the company benefits indirectly from high NAND prices. [WDC-5]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026 / FQ3 FY26	Actual adjusted	3,337.00	1,684.00	1,287.00	WDC-1
Q2 2026 / FQ4 FY26	Actual GAAP	3,747.00	2,028.00	1,563.00	WDC-1
Q2 2026 / FQ4 FY26	Actual adjusted	3,747.00	2,037.00	1,655.00	WDC-1

Numerical anchor: $3,747M sales = $3,335M cloud + $225M client + $187M consumer. Adjusted GP is $2,037M and opex $382M, producing $1,655M OP. GAAP GP is $2,028M and opex $465M; +$9M GP and -$83M opex reconcile the $92M adjusted OP difference. For an actual-quarter product-margin proxy, assume 52% client/consumer GM: $412M x 52% = $214.24M GP, leaving $1,822.76M cloud GP / $3,335M = 54.66%. This fits management description that both categories are now in the 50s. It does not claim the company disclosed those exact segment margins. [WDC-1, WDC-3]

FQ1 FY2027 revenue guidance is $4,100M +/- $100M, adjusted GM 55-56% and opex $390-400M. The model uses the midpoints: $4,100M sales, $2,275.5M GP and $395M opex, yielding $1,880.5M OP (45.87%). The revenue split is $3,650M cloud and $450M client/consumer. At 223 modeled cloud EB, realized cloud price is $16.368/TB. Current guided adjusted EPS of $4.00 +/- $0.15, 17% tax and 388M diluted shares provide a consistency check. They are not substituted for operating profit. [WDC-1, WDC-3]

Balance sheet and capital allocation

July 3 cash was $1,579M against roughly $1.1B debt. The remaining 1.7M SanDisk shares were monetized in the quarter; the model adds no residual SanDisk stake or flash earnings. On September 14, WDC called the remaining $109.505M of 3% convertible notes for November 16 redemption; that is a future event, not already completed debt removal. [WDC-1, WDC-2, WDC-7] The cash-flow cross-check allows capex at 7% of revenue in 2027, 6.5% in 2028, 6% in 2029 and 5.5% in 2030, reflecting September commentary. These are analyst assumptions; normal depreciation remains in vector costs. Buybacks are not converted into an assumed falling denominator.

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis	17	Scaled, qualified head/media platform and few alternatives; Seagate lead prevents monopoly credit
Cycle Stage	9	40TB ramp is underway; HAMR remains a qualification-to-volume transition
TAM Expansion	8	Persistent AI data and wider cloud deployment grow the storage wallet
Catalysts	5	40TB adoption, H1 2027 HAMR start and later performance features
Macro	9	Cloud capex and data retention are strong; construction/power limits affect timing
Alpha	2	Broadly followed; edge comes from separating NAND, HDD and qualification economics
Risks	-4	Four independent risk classes below
Total	46/60 = 77%	High Conviction

46/60 = 77% (High Conviction). The business has a durable infrastructure role, but the company itself is not unavoidable: other qualified HDD suppliers exist. Four distinct risk deductions are applied once. Base is independent of market price, buybacks and the chosen multiple.

4. Quarterly operating model

Four revenue vectors: HAMR cloud, 40TB-class ePMR cloud, other cloud drives, and client/consumer. Potential cloud EB = MIN(capex-derived demand, physical capacity). Allocate potential EB between planned HAMR, higher-density ePMR and residual legacy. Multiply only the planned HAMR units by their explicit qualification/delivery probability; missed HAMR units are not automatically replaced. Normalize a 4% HAMR and 2% advanced-ePMR price/TB premium so shipped cloud EB reproduce the displayed blended ASP. These premiums are analyst assumptions, not disclosed product pricing. Each vector has a separate GM and GP; Reference R&D/SG&A are held in dollars for the incremental pricing branch. UltraSMR is embedded in product economics, never added as another shipment pool.

Revenue vector	Driver and source	Confidence
HAMR cloud drives	WDC-3/5: four qualifications and H1 2027 start; planned EB x conversion factor below x normalized ASP.	MEDIUM timing; LOW mix and product GM
40TB-class ePMR cloud drives	WDC-3: current volume ramp and FQ3 FY27 >50% mix ambition; modeled mix declines as HAMR replaces it.	MEDIUM 2026-27; LOW later
Other cloud drives	WDC-1/3: remaining calibrated cloud pool after newer products; explicit cannibalization.	MEDIUM near term; LOW later
Client and consumer HDDs	WDC-1/3/5: $412M actual; stronger NAND supports pricing but also raises component costs.	MEDIUM Q3 2026; LOW later

Raw cloud capacity rises from 223 EB in Q3 2026 to 318 EB in Q4 2027 and 584 EB in Q4 2030; demand and HAMR qualification can restrict realized shipments. This is a density/automation supply proxy, not announced factory guidance. Reference cloud ASP rises 2.5%, 2.5%, 2%, 2%, 1.5% sequentially through Q4 2027, is flat in 2028, and eases 1% annually in 2029 and 2% in 2030. Active ASP applies the two explicit pricing branches above. Reference HAMR GM ramps to 70%, advanced-ePMR peaks at 64.5% before mix runoff, other cloud settles at 52.5%, and client/consumer at 57%. Later margins are analyst estimates with LOW confidence. Reference opex rises from $395M to $516M quarterly before the 3% incremental pricing-support charge; R&D/SG&A use a 67%/33% allocation informed by reported expense mix, not management segment forecasts. No stock-compensation cash add-back is confused with a lower share count.

Physical and commercial drivers

Quarter	Demand EB	Capacity EB	Shipped EB	Active USD/TB	HAMR share
Q3 2026	239.085	223.000	223.000	16.368	0.00%
Q4 2026	249.480	241.000	241.000	16.876	0.00%
Q1 2027	324.844	263.000	260.370	17.388	4.04%
Q2 2027	324.844	285.000	279.870	17.821	10.39%
Q3 2027	324.844	301.000	294.378	18.254	20.25%
Q4 2027	324.844	318.000	312.435	18.469	33.84%
Q1 2028	389.812	335.000	326.625	18.424	48.72%
Q2 2028	389.812	354.000	342.495	18.378	63.82%
Q3 2028	389.812	378.000	363.825	18.333	74.03%
Q4 2028	389.812	403.000	373.245	18.333	84.33%
Q1 2029	462.162	424.000	404.920	18.291	89.53%
Q2 2029	462.162	446.000	425.484	18.157	91.61%
Q3 2029	462.162	468.000	440.440	18.023	93.70%
Q4 2029	462.162	490.000	440.209	17.888	94.75%
Q1 2030	567.037	512.000	487.552	17.798	95.27%
Q2 2030	567.037	536.000	510.272	17.708	95.80%
Q3 2030	567.037	560.000	532.980	17.619	96.32%
Q4 2030	567.037	584.000	539.536	17.530	96.85%

Demand, capacity, allocation and qualification factors remain as modeled. Active price combines the reference, retained Toshiba concession and scarcity increment. Relative product-price premiums remain normalized to the blended ASP. Neither supplier density nor unmet industry demand becomes extra customer shipments.

Qualification and product allocation

The HAMR factor is 80% in Q1 2027, 85% in Q2, 90% in Q3 and 95% from Q4 onward; this is an analyst qualification-and-delivery reserve applied once to the planned units. Zero HAMR revenue is included in 2026. The stronger weight over time follows the management-stated initial-shipment/ramp sequence, not a claim that future milestones have already occurred. The forward product mix is low-confidence. Existing ePMR/legacy production uses 100% of its planned share. If documented qualification slips, reduce the HAMR factor and its associated exabytes; do not haircut the full company ceiling again. 40TB-class ePMR share is 50% of pre-reserve cloud EB in Q1 2027, consistent with the platform milestone. The high-bandwidth roadmap is not a separate revenue pool.

Quarter	Planned HAMR mix	HAMR factor	40TB ePMR mix	HAMR ship EB	ePMR ship EB	Other cloud EB
Q3 2026	0.0%	100%	25.0%	0.00	55.75	167.25
Q4 2026	0.0%	100%	40.0%	0.00	96.40	144.60
Q1 2027	5.0%	80%	50.0%	10.52	131.50	118.35
Q2 2027	12.0%	85%	60.0%	29.07	171.00	79.80
Q3 2027	22.0%	90%	55.0%	59.60	165.55	69.23
Q4 2027	35.0%	95%	45.0%	105.73	143.10	63.60
Q1 2028	50.0%	95%	35.0%	159.12	117.25	50.25
Q2 2028	65.0%	95%	25.0%	218.59	88.50	35.40
Q3 2028	75.0%	95%	18.0%	269.32	68.04	26.46
Q4 2028	85.0%	95%	12.0%	314.77	46.78	11.69
Q1 2029	90.0%	95%	8.0%	362.52	33.92	8.48
Q2 2029	92.0%	95%	5.0%	389.80	22.30	13.38
Q3 2029	94.0%	95%	3.0%	412.71	13.86	13.86
Q4 2029	95.0%	95%	2.0%	417.10	9.24	13.86
Q1 2030	95.5%	95%	1.0%	464.51	5.12	17.92
Q2 2030	96.0%	95%	0.5%	488.83	2.68	18.76
Q3 2030	96.5%	95%	0.2%	513.38	1.12	18.48
Q4 2030	97.0%	95%	0.0%	522.52	0.00	17.01

HAMR cloud drives

Quarter	Revenue $M	COGS $M	Gross margin	Gross profit $M
Q3 2026	0.000	0.000	58.00%	0.000
Q4 2026	0.000	0.000	58.00%	0.000
Q1 2027	188.040	76.452	59.34%	111.587
Q2 2027	530.087	204.315	61.46%	325.771
Q3 2027	1,109.965	399.284	64.03%	710.682
Q4 2027	1,985.885	677.457	65.89%	1,308.428
Q1 2028	2,969.768	986.320	66.79%	1,983.449
Q2 2028	4,053.695	1,309.585	67.69%	2,744.110
Q3 2028	4,969.320	1,560.140	68.60%	3,409.180
Q4 2028	5,791.698	1,760.960	69.60%	4,030.737
Q1 2029	6,646.815	1,984.481	70.14%	4,662.334
Q2 2029	7,093.413	2,124.628	70.05%	4,968.785
Q3 2029	7,451.718	2,239.265	69.95%	5,212.453
Q4 2029	7,473.148	2,253.210	69.85%	5,219.938
Q1 2030	8,280.681	2,496.688	69.85%	5,783.993
Q2 2030	8,669.450	2,613.904	69.85%	6,055.545
Q3 2030	9,057.660	2,730.953	69.85%	6,326.707
Q4 2030	9,171.071	2,765.147	69.85%	6,405.924

40TB-class ePMR cloud drives

Quarter	Revenue $M	COGS $M	Gross margin	Gross profit $M
Q3 2026	926.119	388.970	58.00%	537.149
Q4 2026	1,646.221	671.952	59.18%	974.270
Q1 2027	2,305.292	903.079	60.83%	1,402.213
Q2 2027	3,058.192	1,148.642	62.44%	1,909.550
Q3 2027	3,023.944	1,102.611	63.54%	1,921.333
Q4 2027	2,635.979	938.104	64.41%	1,697.875
Q1 2028	2,146.169	755.093	64.82%	1,391.076
Q2 2028	1,609.611	567.713	64.73%	1,041.898
Q3 2028	1,231.265	441.443	64.15%	789.822
Q4 2028	844.134	302.646	64.15%	541.488
Q1 2029	609.964	221.376	63.71%	388.588
Q2 2029	397.998	144.952	63.58%	253.045
Q3 2029	245.523	90.968	62.95%	154.556
Q4 2029	162.425	60.399	62.81%	102.026
Q1 2030	89.517	33.288	62.81%	56.229
Q2 2030	46.616	17.335	62.81%	29.281
Q3 2030	19.380	7.207	62.81%	12.174
Q4 2030	0.000	0.000	63.00%	0.000

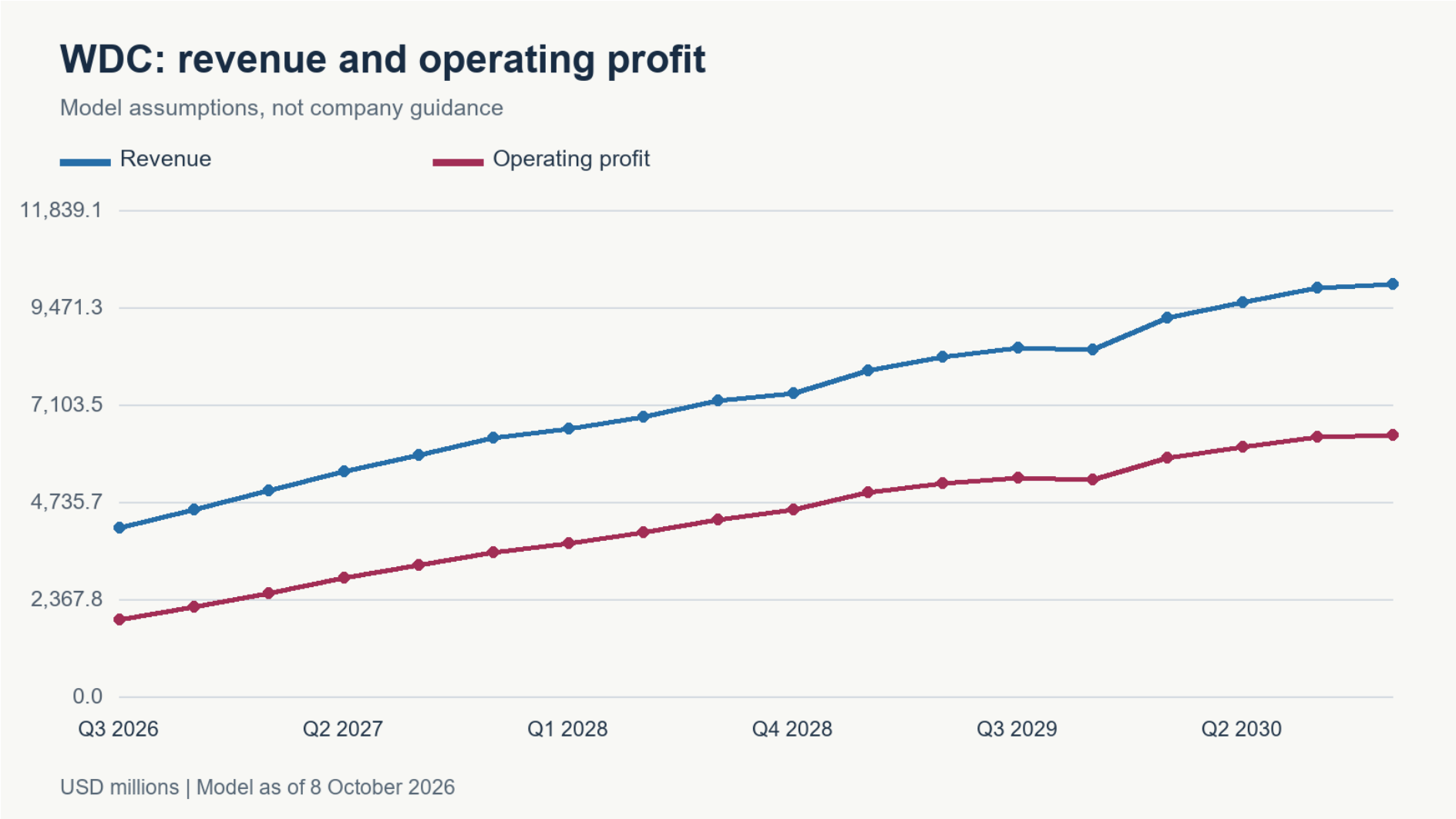
Other cloud drives

Quarter	Revenue $M	COGS $M	Gross margin	Gross profit $M
Q3 2026	2,723.881	1,228.530	54.90%	1,495.351
Q4 2026	2,420.914	1,072.398	55.70%	1,348.515
Q1 2027	2,034.082	887.357	56.38%	1,146.724
Q2 2027	1,399.173	608.152	56.53%	791.020
Q3 2027	1,239.763	543.183	56.19%	696.580
Q4 2027	1,148.575	510.397	55.56%	638.177
Q1 2028	901.752	406.147	54.96%	495.605
Q2 2028	631.220	288.121	54.35%	343.099
Q3 2028	469.436	217.130	53.75%	252.306
Q4 2028	206.896	96.721	53.25%	110.175
Q1 2029	149.501	70.545	52.81%	78.956
Q2 2029	234.116	110.893	52.63%	123.223
Q3 2029	240.709	114.459	52.45%	126.251
Q4 2029	238.860	114.029	52.26%	124.832
Q1 2030	307.166	146.637	52.26%	160.529
Q2 2030	319.913	152.722	52.26%	167.191
Q3 2030	313.506	149.664	52.26%	163.842
Q4 2030	287.086	137.051	52.26%	150.035

Client and consumer

Quarter	Revenue $M	COGS $M	Gross margin	Gross profit $M
Q3 2026	450.000	207.000	54.00%	243.000
Q4 2026	480.000	216.000	55.00%	264.000
Q1 2027	475.000	211.375	55.50%	263.625
Q2 2027	490.000	215.600	56.00%	274.400
Q3 2027	500.000	217.500	56.50%	282.500
Q4 2027	515.000	221.450	57.00%	293.550
Q1 2028	505.000	217.150	57.00%	287.850
Q2 2028	515.000	218.875	57.50%	296.125
Q3 2028	525.000	223.125	57.50%	301.875
Q4 2028	540.000	226.800	58.00%	313.200
Q1 2029	525.000	223.125	57.50%	301.875
Q2 2029	535.000	227.375	57.50%	307.625
Q3 2029	545.000	234.350	57.00%	310.650
Q4 2029	560.000	240.800	57.00%	319.200
Q1 2030	540.000	232.200	57.00%	307.800
Q2 2030	550.000	236.500	57.00%	313.500
Q3 2030	560.000	240.800	57.00%	319.200
Q4 2030	575.000	247.250	57.00%	327.750

Consolidated operating forecast



WDC: revenue and operating profit. USD millions. Model assumptions, not company guidance

Quarter	Revenue	COGS	GP	R&D	SG&A	Other opex	OP	OP margin
Q3 2026	4,100.000	1,824.500	2,275.500	264.650	130.350	0.000	1,880.500	45.87%
Q4 2026	4,547.135	1,960.350	2,586.785	268.000	132.000	0.717	2,186.068	48.08%

Quarter	Revenue	COGS	GP	R&D	SG&A	Other opex	OP	OP margin
Q1 2027	5,002.413	2,078.264	2,924.150	271.350	133.650	1.500	2,517.649	50.33%
Q2 2027	5,477.451	2,176.710	3,300.741	274.700	135.300	2.354	2,888.388	52.73%
Q3 2027	5,873.673	2,262.578	3,611.095	278.050	136.950	3.208	3,192.886	54.36%
Q4 2027	6,285.438	2,347.408	3,938.031	281.400	138.600	3.329	3,514.701	55.92%
Q1 2028	6,522.689	2,364.710	4,157.979	286.090	140.910	3.481	3,727.498	57.15%
Q2 2028	6,809.526	2,384.295	4,425.232	290.780	143.220	3.650	3,987.582	58.56%
Q3 2028	7,195.022	2,441.838	4,753.183	295.470	145.530	3.877	4,308.306	59.88%
Q4 2028	7,382.727	2,387.127	4,995.600	300.160	147.840	3.977	4,543.623	61.54%
Q1 2029	7,931.280	2,499.527	5,431.754	305.520	150.480	3.249	4,972.505	62.69%
Q2 2029	8,260.527	2,607.849	5,652.678	310.880	153.120	2.285	5,186.394	62.79%
Q3 2029	8,482.951	2,679.041	5,803.910	316.240	155.760	1.187	5,330.723	62.84%
Q4 2029	8,434.434	2,668.439	5,765.995	321.600	158.400	0.000	5,285.995	62.67%
Q1 2030	9,217.365	2,908.813	6,308.552	327.630	161.370	0.000	5,819.552	63.14%
Q2 2030	9,585.978	3,020.461	6,565.517	333.660	164.340	0.000	6,067.517	63.30%
Q3 2030	9,950.546	3,128.623	6,821.923	339.690	167.310	0.000	6,314.923	63.46%
Q4 2030	10,033.158	3,149.448	6,883.709	345.720	170.280	0.000	6,367.709	63.47%

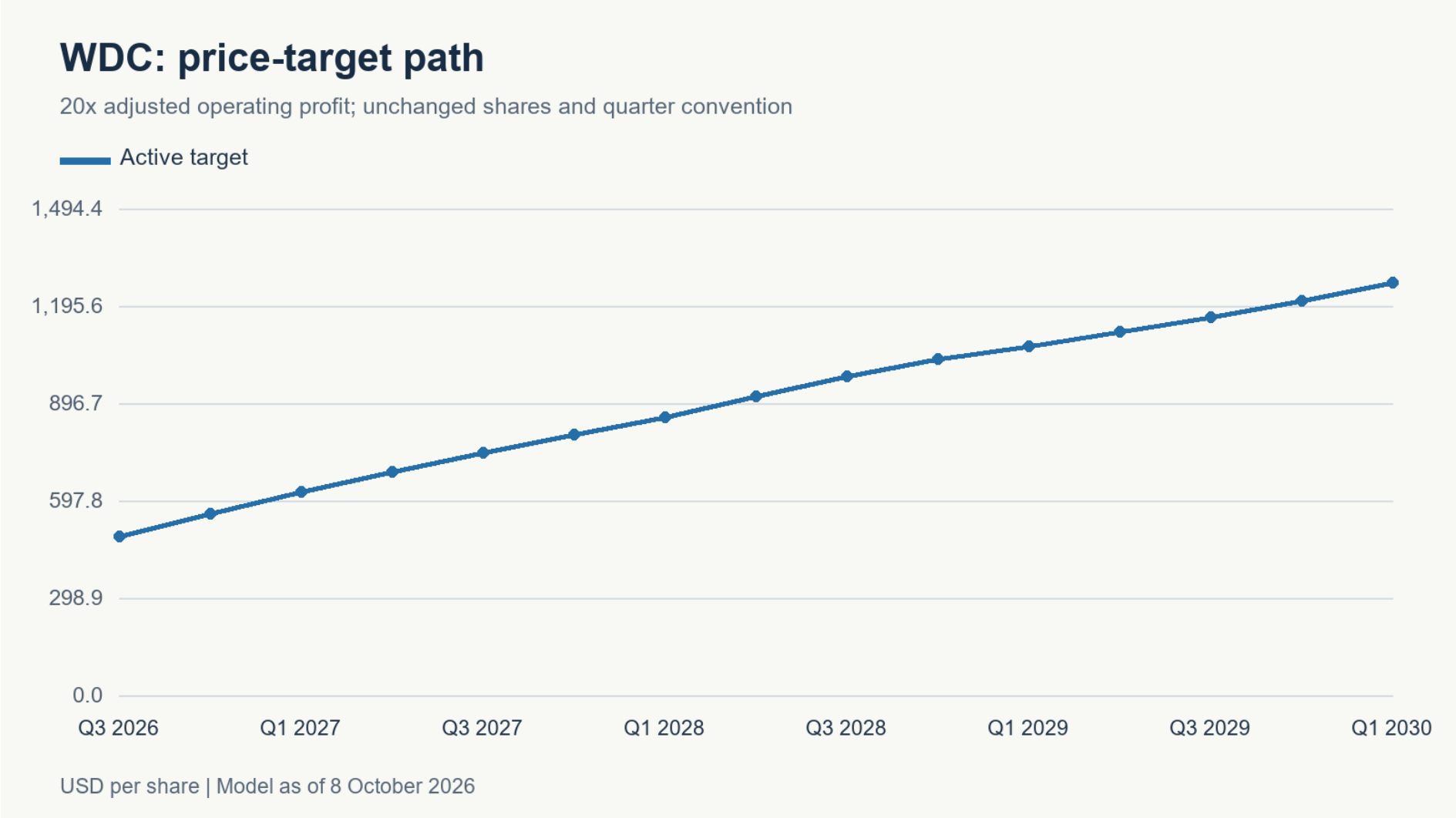
All forecast quarters are analyst estimates. Ordinary production depreciation remains in COGS; the sanctioned adjusted basis is retained. Pricing scenarios enter revenue/costs once, with no extra probability applied to the target.

Calendar and fiscal reconciliation

Calendar year	Coverage	Revenue	GP	OP	OP margin
2026	2 quarters	8,647.135	4,862.285	4,066.568	47.03%
2027	4 quarters	22,638.976	13,774.016	12,113.625	53.51%
2028	4 quarters	27,909.964	18,331.994	16,567.009	59.36%
2029	4 quarters	33,109.192	22,654.337	20,775.616	62.75%
2030	4 quarters	38,787.047	26,579.701	24,569.701	63.35%

Fiscal year ending around June	Revenue	OP
FY2027	19,127.000	9,472.605
FY2028	25,491.326	14,422.668
FY2029	30,769.556	19,010.828
FY2030	35,720.728	22,503.787

5. Forward-four-quarter ceiling derivation



WDC: price-target path. USD per share. 20x adjusted operating profit; unchanged shares and quarter convention

Ceiling quarter	Earnings window	F4Q OP $M	Multiple	Diluted shares M	Ceiling USD
Q3 2026	Q3 2026 to Q2 2027	9,472.605	20.0x	388.000	488.278597
Q4 2026	Q4 2026 to Q3 2027	10,784.991	20.0x	388.000	555.927384

Ceiling quarter	Earnings window	F4Q OP $M	Multiple	Diluted shares M	Ceiling USD
Q1 2027	Q1 2027 to Q4 2027	12,113.625	20.0x	388.000	624.413639
Q2 2027	Q2 2027 to Q1 2028	13,323.473	20.0x	388.000	686.776980
Q3 2027	Q3 2027 to Q2 2028	14,422.668	20.0x	388.000	743.436487
Q4 2027	Q4 2027 to Q3 2028	15,538.088	20.0x	388.000	800.932350
Q1 2028	Q1 2028 to Q4 2028	16,567.009	20.0x	388.000	853.969548
Q2 2028	Q2 2028 to Q1 2029	17,812.016	20.0x	388.000	918.145145
Q3 2028	Q3 2028 to Q2 2029	19,010.828	20.0x	388.000	979.939570
Q4 2028	Q4 2028 to Q3 2029	20,033.244	20.0x	388.000	1,032.641453
Q1 2029	Q1 2029 to Q4 2029	20,775.616	20.0x	388.000	1,070.908062
Q2 2029	Q2 2029 to Q1 2030	21,622.664	20.0x	388.000	1,114.570284
Q3 2029	Q3 2029 to Q2 2030	22,503.787	20.0x	388.000	1,159.989009
Q4 2029	Q4 2029 to Q3 2030	23,487.987	20.0x	388.000	1,210.720991
Q1 2030	Q1 2030 to Q4 2030	24,569.701	20.0x	388.000	1,266.479455

For each row, F4Q OP is the sum of four consecutive operating quarters beginning at the named quarter. Ceiling = F4Q OP × 20 / that quarter's diluted-share denominator. No separate net-income, EBITDA or enterprise-value method replaces the sanctioned basis.

Q4 2027 worked example: (3,514.701193 + 3,727.498245 + 3,987.581936 + 4,308.306219) × 20 / 388.000000 = **800.932350 USD/share**.

Additional shares	Q4 2027 ceiling
0.00%	800.9324
5.00%	762.7927
10.00%	728.1203

6. Cross-book relationships and monitoring

WDC <-> STX: compare density, EB, net pricing, customer acceptance and manufacturing yields. Higher industry demand helps both; one supplier gaining share must be reconciled with the other supplier model.

WDC <-> SNDK / Kioxia / Samsung / SK hynix / Micron: storage tiers share AI demand. High NAND prices help HDD economics in some uses but raise DRAM/NAND input cost inside the drive. WDC receives none of the earlier NAND +23%/+20%/+15% forecast as a direct ASP multiplier. SanDisk separation means no NAND earnings are consolidated.

WDC <-> optical / compute holdings: deployments create persistent data, but common hyperscaler capex also creates correlated customer risk. Do not add full optical, compute and memory TAMs into HDD sales.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Technology / qualification execution	-1	HAMR reliability, yields or 40TB adoption can delay shipments and increase costs.
Demand and buyer concentration	-1	Large customers can defer data-center deployment, digest inventory or negotiate the value split.
Structural competition and substitution	-1	Seagate/Toshiba density gains, cheap NAND or performance-oriented SSD migration can limit returns.
Financial, input-cost and policy exposure	-1	Higher manufacturing capex, memory inputs, convert settlement, export limits or geographic disruption can erode cash generation.

Dated triggers and counters

The next issuer results and 2027 new-order renewals must reconcile realized price per TB and shipped exabytes to the reported spot-market opportunity. A high distributor spot quote without manufacturer net realization reduces the pricing branch, even when demand remains strong.

Checkpoint	Positive evidence	Counter / model action
1. Next quarterly release / Q4 2026	Revenue within current guide and GM near 55-56%	Below $4.0B or 55% requires a mix/pricing/cost bridge.
2. Q1 2027	40TB ePMR around half of nearline EB; initial qualified HAMR shipments	Cut modeled HAMR conversion if qualification does not become accepted revenue.
3. Q2-Q4 2027	HAMR scales and UltraSMR approaches fiscal-year-end target	Keep overlapping technology shares distinct; missed yields reduce capacity and margin.
4. November 16, 2026 / next filing	Convertible settlement and reported diluted shares reconcile	Update the denominator only on actual or guided settlement economics.
5. 2028 and each subsequent year	Capex, EB and retained-data demand support the modeled growth	Lower supply/ASP assumptions if customer inventory or competing capacity rises; no second ceiling probability haircut.

Consolidated near-term guidance is HIGH confidence; product-level profit allocation and the cloud-only EB split are proxies, not disclosures. Beyond 2027 all operating extensions are LOW confidence. Current Yahoo high-revenue-estimate tables could not be retrieved, so that cross-check is unavailable and no stale snippet is treated as a floor. HAMR probabilities apply once at revenue. The artifact is a single base case, not a claim that every announced long-range capacity or performance goal is delivered on time.

8. Sources and snapshot

- **HDD-P1** - Supplied Morgan Stanley channel-check crop HT9RXOCXIAAfTJQ.jpg, reviewed October 8, 2026. October 5 date is inferred from the exact-match secondary synopsis; crop is undated. Original pixels embedded above. Deal sample and full broker report remain unavailable.
- **WDC-1** - August 5, 2026. FY2026 Q4 release and financial reconciliation. Latest actuals, prior-quarter comparison and FQ1 FY2027 guidance.
- **WDC-2** - August 14, 2026. FY2026 Form 10-K. Standalone HDD perimeter, competitors, share snapshot and risks.
- **WDC-3** - August 5, 2026. FY2026 Q4 management call. Latest earnings-call operating milestones, end-market revenue and EB.
- **WDC-4** - April 30, 2026. FY2026 Q3 management call. Second latest call: pricing mechanics, qualification and manufacturing lead times.
- **WDC-5** - September 9, 2026. Citi conference - Western Digital management. More recent HAMR timing, cost headwinds and capex commentary.
- **WDC-6** - September 21, 2026 close. WDC dated market snapshot. Header price only; does not determine Base or ceilings.
- **WDC-7** - September 14, 2026. Convertible redemption notice - Form 8-K. November 16 planned redemption, principal outstanding and conversion terms.
- **WDC-8** - September 15, 2026. WD AI storage research update. Company-sponsored survey supports direction of data growth; not added as a separate sales forecast.
- **D1** - July 30, 2026. Amazon Q2 management call. Management cash-capex outlook: $220B in CY2026; increase partly reflects memory cost inflation.
- **D2** - July 29, 2026. Microsoft FY2026 Q4 management call. CY2026 capex about $175B after lease classification change; physical investment expectations unchanged.
- **D3** - July 22, 2026. Alphabet Q2 management call. CY2026 capex $195-205B; use $200B midpoint.
- **D4** - July 29, 2026. Meta Q2 results. CY2026 capex including principal finance-lease payments $130-145B; use $137.5B midpoint.

Model snapshot: October 8, 2026; latest reported earnings and market quote retain their original dates. WDC. Framework 8.0. Base 46/60 (77%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/WDC_WesternDigital_DeepDive.pdf and .md. The companion company worksheet uses this same model and links its ceiling series to the master portfolio.